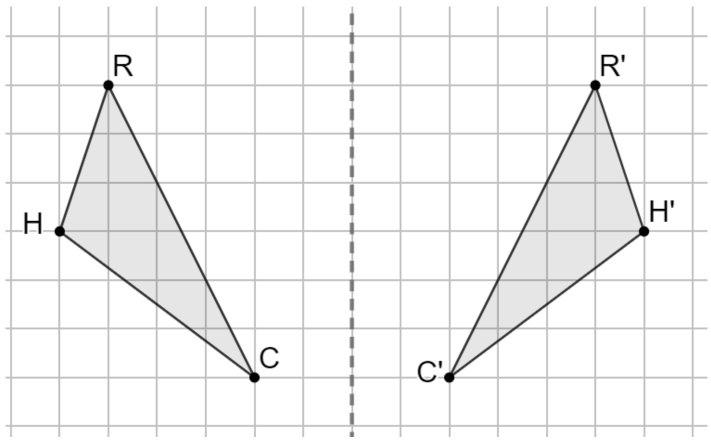
Geometry
Unit 4
Lesson 6 Practice

Name_Answer Key________
Date___________________
Period_________________

Triangles RHC and $R'H'C'$ are shown on grid paper.

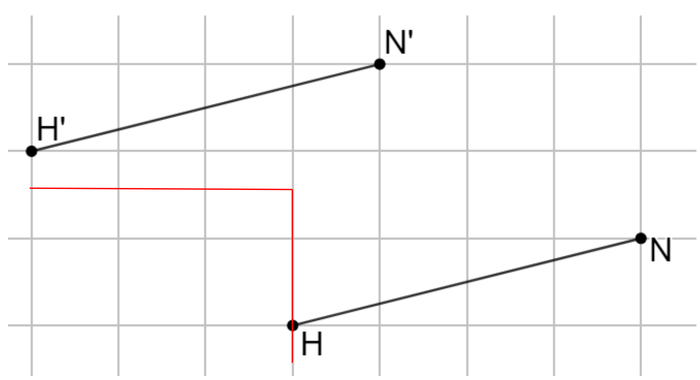


1. Explain what method(s) can be used to determine if each angle of $\triangle RHC$ has the same measure as $\triangle R'H'C'$.
Patty paper can be folded over the line of reflection to map each angle of $\triangle RHC$ to $\triangle R'H'C'$.

2. Explain what method(s) can be used to determine if each side of $\triangle RHC$ has the same length as $\triangle R'H'C'$.
Patty paper can be folded over the line of reflection to map each side of $\triangle RHC$ to $\triangle R'H'C'$.

3. Explain why the transformation of $\triangle RHC$ shown on the grid can be classified as isometry.
Because both the angle measures and side lengths are preserved.

Line segments HN and $H'N'$ are shown on the grid.



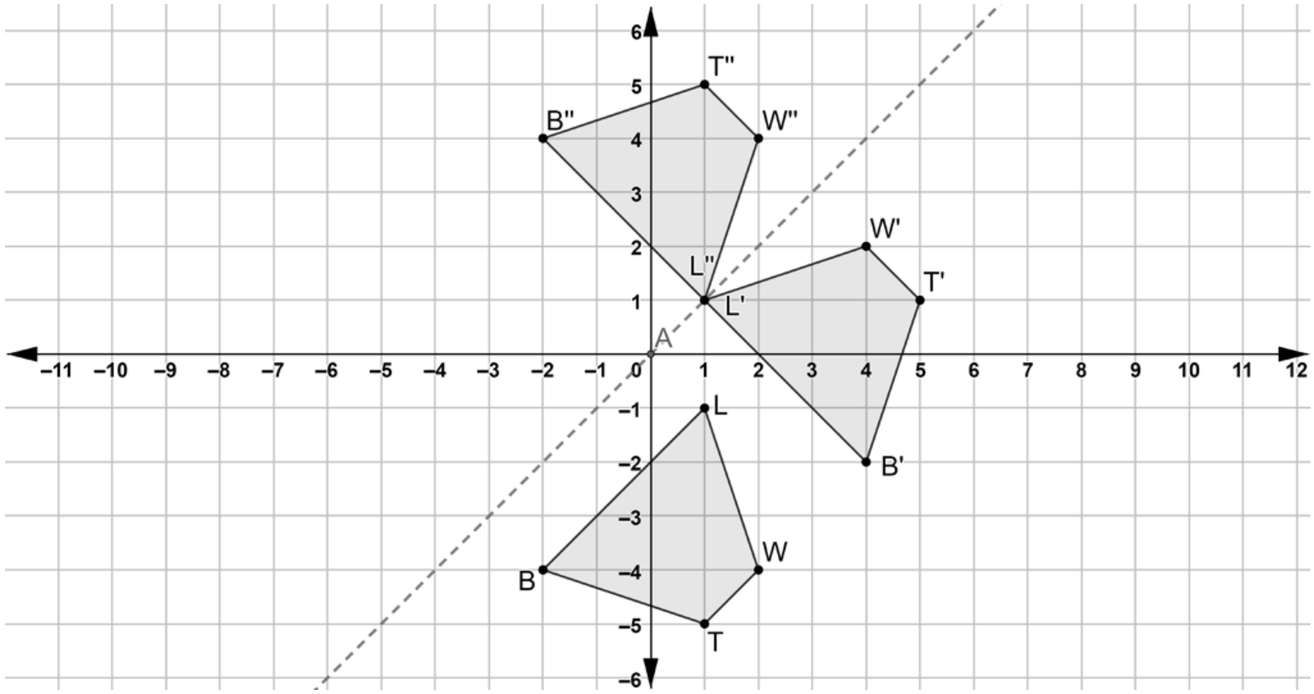
4. Show the mapping of point H to H'.

5. Describe the transformation applied to $\overline{HN}$ that resulted in $\overline{H'N'}$.
$(x, y) \rightarrow (x - 3, y + 2)$

6. Complete the statement.

o Each point on o Only the endpoints of	$\overline{HN}$ can be mapped to	o a corresponding point on o the endpoints of	$\overline{H'N'}$.

Quadrilaterals $LWTB$, $L'W'T'B'$, and $L''W''T''B''$ are shown on the coordinate grid.



7. Describe what movements should be done with a patty paper copy of quadrilateral $LWTB$ so that the copy of $LWTB$ lays on top of $L'W'T'B'$.
Rotate 90° counterclockwise about the origin.

$(x, y) \rightarrow (-y, x)$

8. Describe what movements should be done with a patty paper copy of quadrilateral $L'W'T'B'$ so that the copy of $L'W'T'B'$ lays on top of $L''W''T''B''$.

Reflect over $y = x$
$(x, y) \rightarrow (y, x)$

9. Complete the statements.

The transformations used in both sequences

- o are
- o are not

examples of isometry. All of the points of

- o $L'W'T'B'$
- o $L''W''T''B''$
- o $L'W'T'B'$ and $L''W''T''B''$

can be mapped to all of the points of $LWTB$.

10. Use the definition of congruence in terms of rigid motions to explain how both sequences of transformations for quadrilateral $LWTB$ resulted in $LWTB \cong L''W''T''B''$.
Rigid motion transformations result in congruent figures.